



Cubsat 12U | Launch: Q1 2027 | Distant Retrograde orbit in Deep Space

Objectives

Space Weather Operations

Providing a **near real time monitoring** of the deep space environment well beyond L1

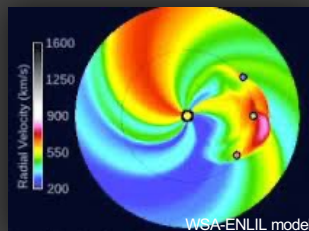
Giving **real time alerts** of solar energetic particles (SEPs) and interplanetary perturbations (**10 times improvement in the lead time of geoeffective events prediction**)



Space Weather Science

Giving useful scientific data.

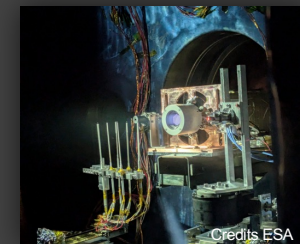
- acceleration and transport of energetic particles;
- propagation models of interplanetary perturbations and energetic particles;
- large-scale turbulent eddies in the solar wind (in combination with data from L1 spacecraft);
- extension of the magnetotail.

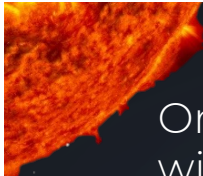


Expanding the boundaries of space technology

Flying for the first time ever the unexplored DRO, which have a critical importance also in other space contexts

Demonstrate reliability, availability and maintainability of the CubeSat technologies in deep space mission



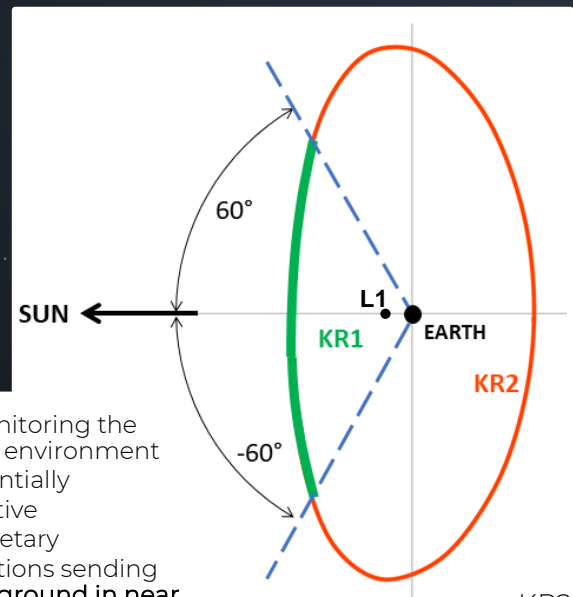


Orbiting the DRO, HENON will remain for a long time upstream of the Earth at a distance much larger than L1, — around 0.1 AU.



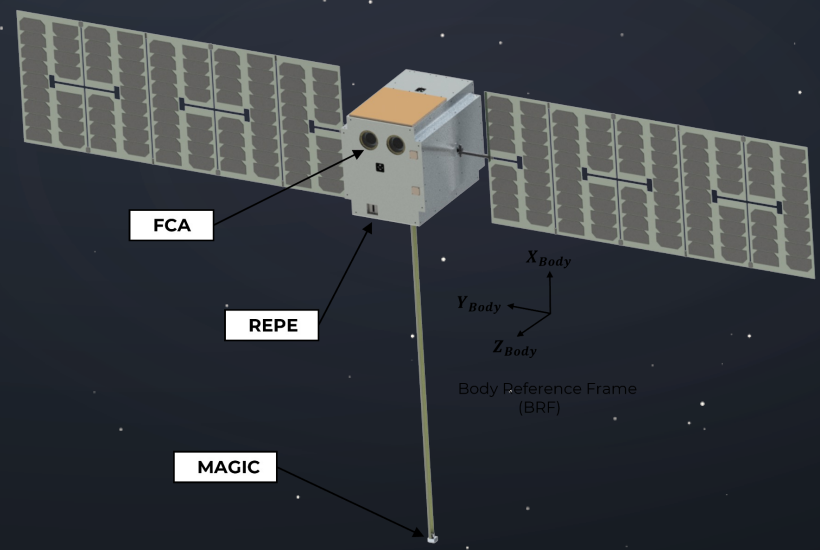
Payload

REPE ASRO Turku Univ, Finland	Energetic proton and electron fluxes	Protons energy range 3-200+ MeV, $\Delta E/E \leq 50\%$ at energies below 70 MeV and $\Delta E/E \leq 80\%$ at higher energies; electrons energy range 0.25-10.4 MeV, $\Delta E/E \leq 40\%$ at energies below 1 MeV and $\Delta E/E \leq 80\%$ at higher energies; azimuth angular resolution of 45° @ 0.03 Hz
MAGIC Imperial College UK	Magnetic field components	± 2000 nT with a digital resolution of 2pT* and noise spectral density < 500 pT/sqrt(Hz) at 1 Hz @ 10 Hz
FCA CharlesUniversity Czech Republic	1D ion and alpha particle energy distribution and ion moments	E/q in 100–4000 eV range; Velocity: 200 – 1000 km/s, Density: 0.1 – 200 cm ⁻³ , Temperature: 0.1 – 100 eV @ 0.2 Hz



KR1 - monitoring the radiation environment and potentially geoeffective interplanetary perturbations sending alerts to ground in near real-time

KR2 - Advanced measurement



Reserve

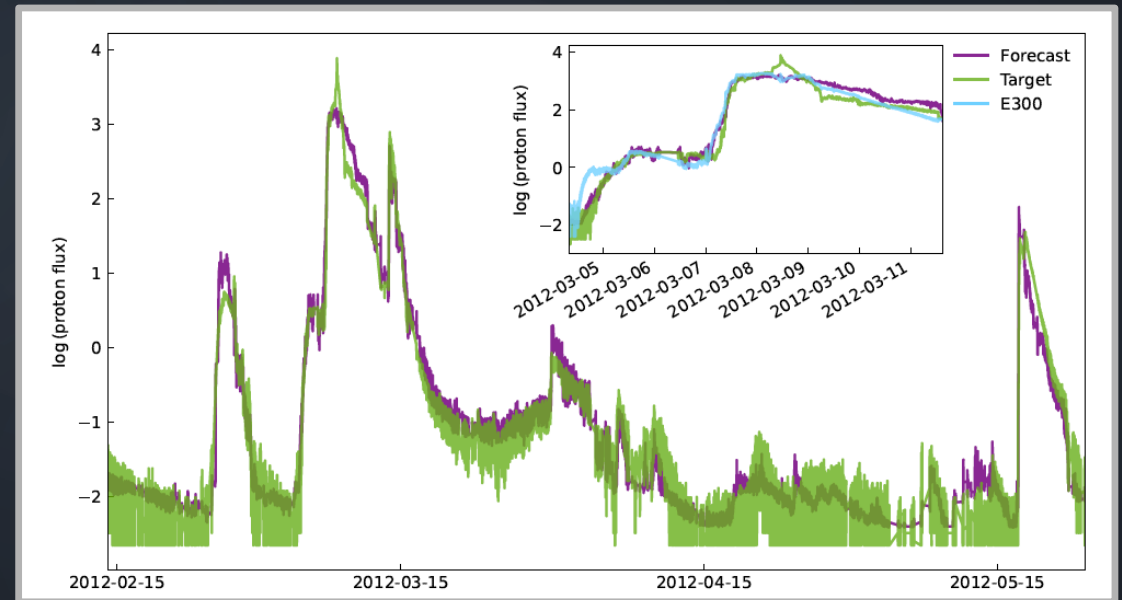
HENON Alerts

SEP alert model for on board operations

Reliable prediction of the >10 MeV proton flux time series with an advance of 1 hour (i.e., before an increase of the proton flux is directly measured) by taking as input only the electron fluxes.

The algorithm ensures:

- maximization of the warning time;
- good computational efficiency;
- good performance for the SEP alert system.



R^2	RMSE	Warning Time [min]
0.82	0.38	54

HENON Alerts

Geomagnetic storms alert model for on board operations - ANN Model

The model predict the Sym-H index 1 hour ahead every 20 minutes from only interplanetary magnetic field data (as measured at L1).

Geomagnetic storms alerts for on board operations – Threshold method

Based on the overcome for three hours of thresholds of specific interplanetary parameters. The thresholds used are 4 mV/m and -6 nT for the VBz and the Bz, respectively. The performance of the method is very good (POD= 86% and a FAR=14%).

